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POWER SUPPLY
With Digital Volt-Amp Panel Meter

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This is a variable power supply with digital volt-amp panel meter. Its advantage is that you do not need a separate voltmeter, ammeter, or multimeter while using this variable power supply as the digital panel meter shows the voltage as well as current.

The circuit diagram of the power supply with digital VA panel meter is shown in Fig. 2.

The circuit comprises LM317 adjustable voltage regulator (IC1), rectifier diodes (D1-D4), digital panel meter (DPM1), 230V AC primary to 15V, 1A secondary transformer (X1), and an LED, and a few other components.

LM317 is a popular adjustable voltage regulator that comes with overvoltage protection, internal current limiting, and overload protection. So, it provides a regulated and stabilised voltage at the output, irrespective of any fluctuation in input voltage and load current. Here, the IC is designed to accept 15V as input and produce a variable DC output voltage up to 15V.

SPECIFICATIONS OF DIGITAL VOLT-AMP PANEL METER

Operating voltage	: DC 4.5~30V
Operating current	: 20mA
Display colour	: Dual-colour (red and blue)
Dimensions	: 48×29×21mm
Refresh rate	: About 500ms
Measure accuracy	: 1%
Working current	: 20mA
Measuring range	: DC 0-100V, 0-10A
Operating temperature	: 10 to 65°C

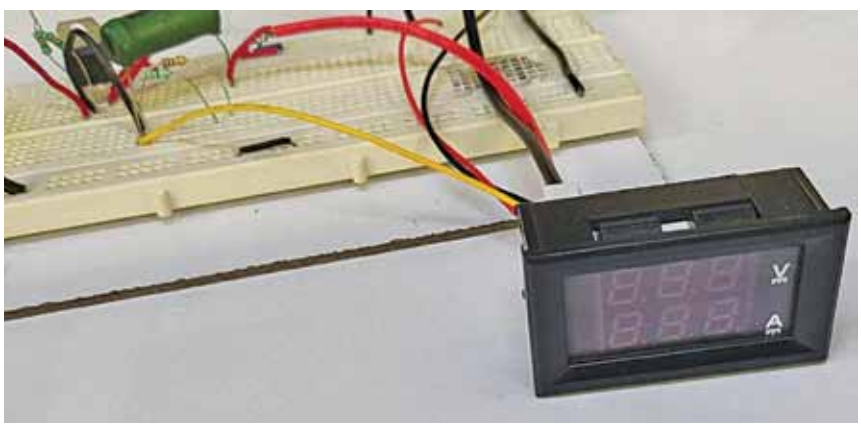

Fig. 1: Author's prototype wired on a breadboard

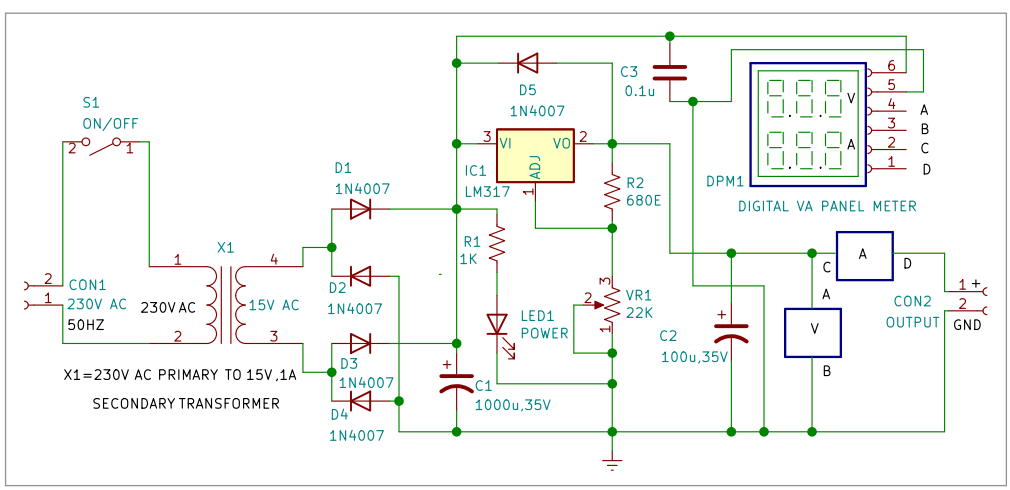

Fig. 2: The circuit diagram

Power supply for the circuit is built around transformer X1, four diodes (D1-D4) that form a bridge rectifier, and filter capacitors C1 and C2. The secondary terminals of the transformer are connected to bridge rectifier diodes D1-D4. The glowing of LED1 indicates presence of power supply across the transformer.

Pot VR1 is connected to variable pin of the regulator IC for voltage adjustment. Capacitor C3 is used for compensating transient currents. Diode D5 connected across pins 2 and 3 of the voltage regulator gives short-circuit protection. It prevents external capacitor from discharging through the IC during an input short-circuit. The regulator maintains a constant voltage of desired level at the output. A suit-